

UPS Delivery Trucker Crash Risk Predictor

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Problem Statement

A Data-Driven Analytic Framework for Predicting UPS Delivery Trucker Crash Risk

Niche – No real framework out there for a driver to evaluate the risk involved when making a delivery!

- *Average 1300 reported accidents per year*
- *This is for the truckers!*
- *For a driver to easily predict High/Moderate/Low risk of delivery*



Literature Review

Research Focus:

- The occurrence of UPS delivery crashes across the United States
- The Severity of the crashes
- How preventable are they? To what extent?
- How predictable are these crashes?
- To what extent can we prevent them?
- To what extent can we predict them?



Literature Review

Method	Purpose	Example Studies
Multinomial logit	Severity heterogeneity	Okafor et al., 2022
Bayesian networks	Dependency modeling	Wu et al., 2023
Machine learning	Crash risk classification	Liu et al., 2018
Exposure-based modeling	Context-sensitive risk	Hosseinpour & Haleem, 2022

Risk Domain	Key Findings	Representative Studies
Fatigue & workload	Time-on-task increases crash risk; rest timing matters	Hanowski et al., 2009; Chen & Xie, 2014a
Operational pressure	Tight schedules elevate unsafe behaviors	Chen et al., 2021
Last-mile exposure	Dense routes raise injury and crash risk	Iacobucci et al., 2023
Compensation design	Incentives influence fatigue and risk	Rodríguez et al., 2006; Mahajan et al., 2019

Research Methodology

- **Predictive + Decision-Support Framework**
- **Data-Driven Modeling**
- **From Model → Real-World Application**

Data Acquisition
 **FMCSA**
Federal Motor Carrier Safety Administration



Physical Research

Industrial Visits!

- Visited 4 UPS outlets across Massachusetts
- East Longmeadow, Agawam, Northampton & Ludlow
- Surveys and Insights from Managers
- *Direction for data collection*



The Data

Source : data.transport.gov

Datasets :

- **crash_file.csv** (Let's call it **df**)
- **sms_input.csv** (Let's call it **df1**)

```
1 print(df.shape)
  print(df1.shape)

(6525, 59)
(2311, 22)
```

df has-

- 6525 rows
- 59 columns

df1 has-

- 2311 rows
- 22 columns

But how many of these columns could we really use?

Data Cleaning

Operations performed:

- Handle date and time
- Handle structure – Category/quantitative
- Deal with missing values
- Deal with "Unknowns"

OBSERVATIONS

Completely Useless columns:

- crash_carrier_city_code
- crash_colonia

Best Clean Predictors:

- Weather
- Road surface
- Trafficway
- Vehicle type

Type	Action (later)
100% missing	Drop
>90% missing	Likely drop
40–75%	Convert to "Unknown"
<15%	Keep + minor handling

Classifying outcomes for driver

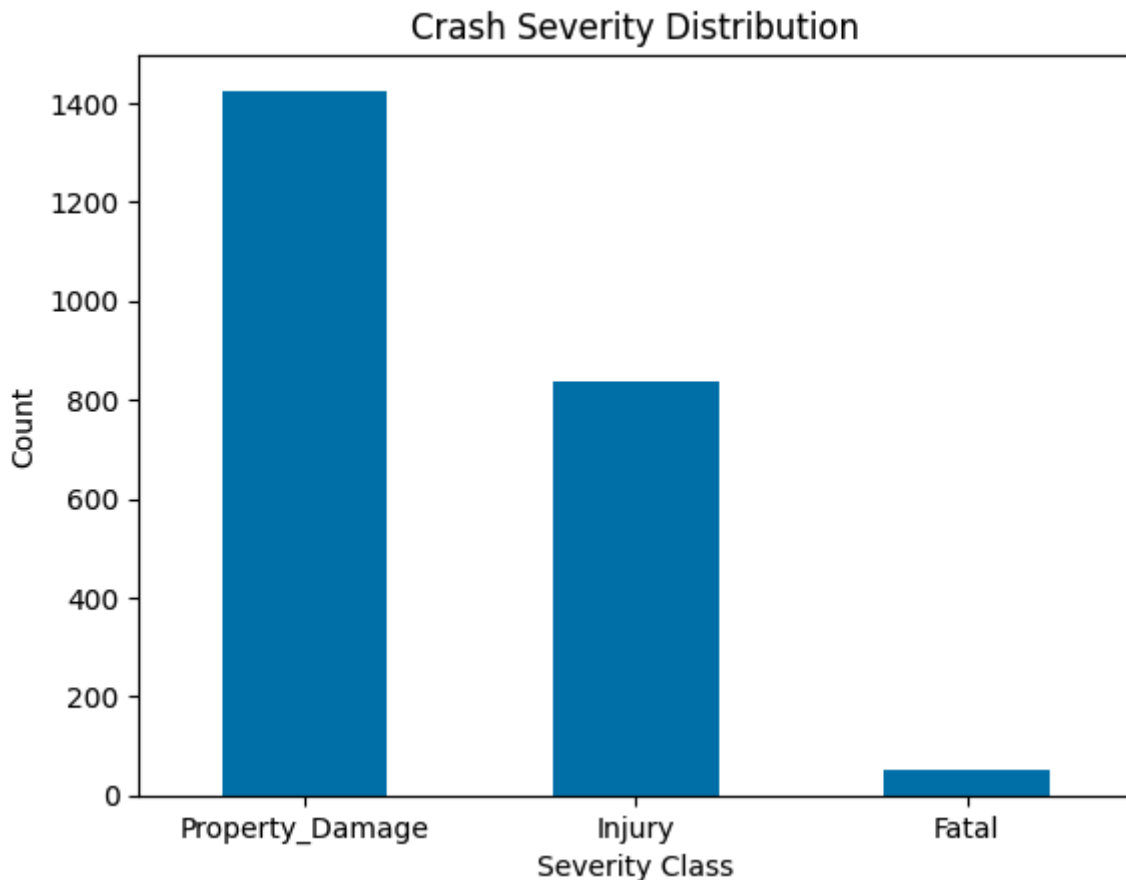
Driver needs to know:

- Do I risk ***Property Damage***?
- Am I likely to get ***Injured***?
- Is it outright ***Fatal***?

	count
severity_class	
Property_Damage	1425
Injury	835
Fatal	51

dtype: int64

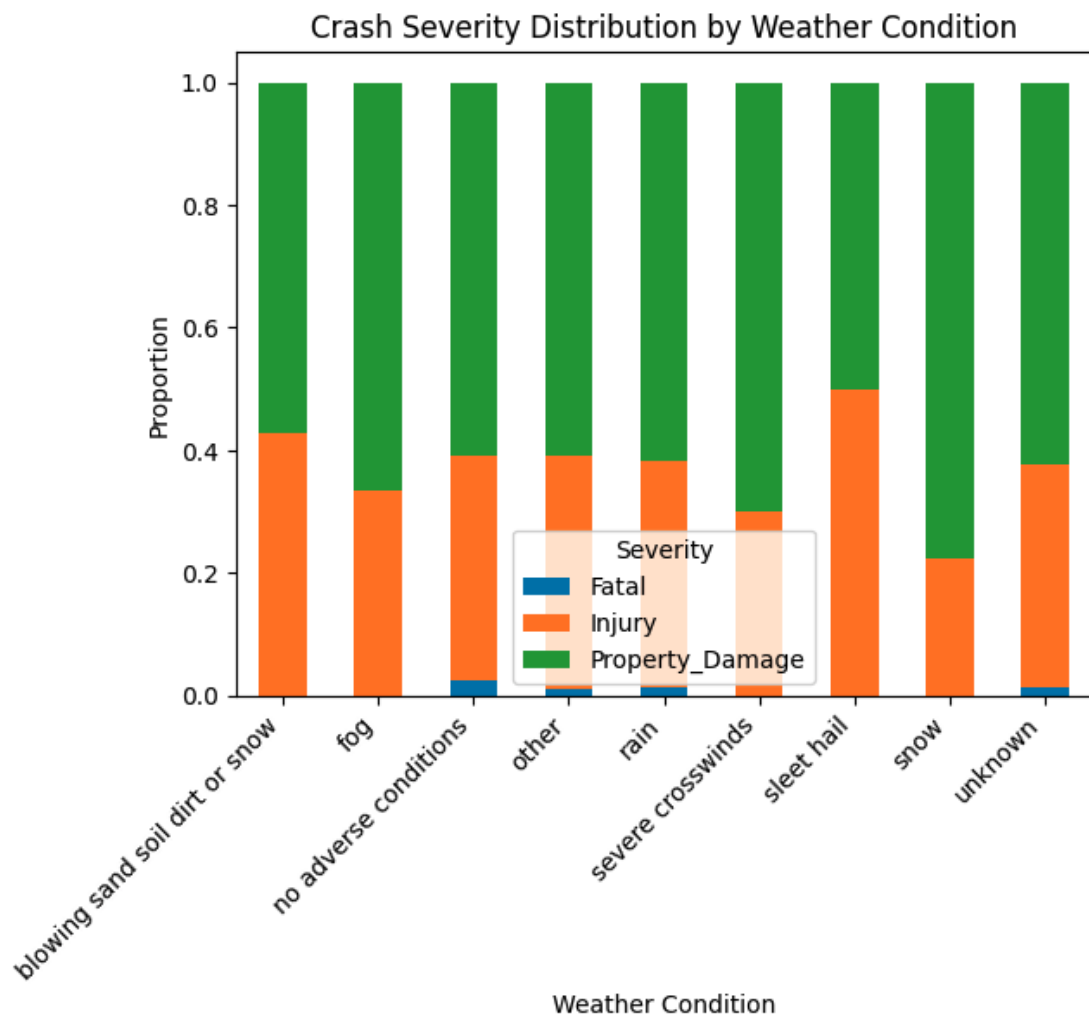
Exploratory Visualizations - 1



- **Majority of crashes are non-severe (PDO dominates)**
Over 60% are property-damage-only, while fatal crashes are extremely rare (~2%)
- **Clear class imbalance in the data**
Injury crashes (~35%) form a meaningful middle category, but fatal events are rare yet high-impact
- **Justifies multi-class severity modeling**

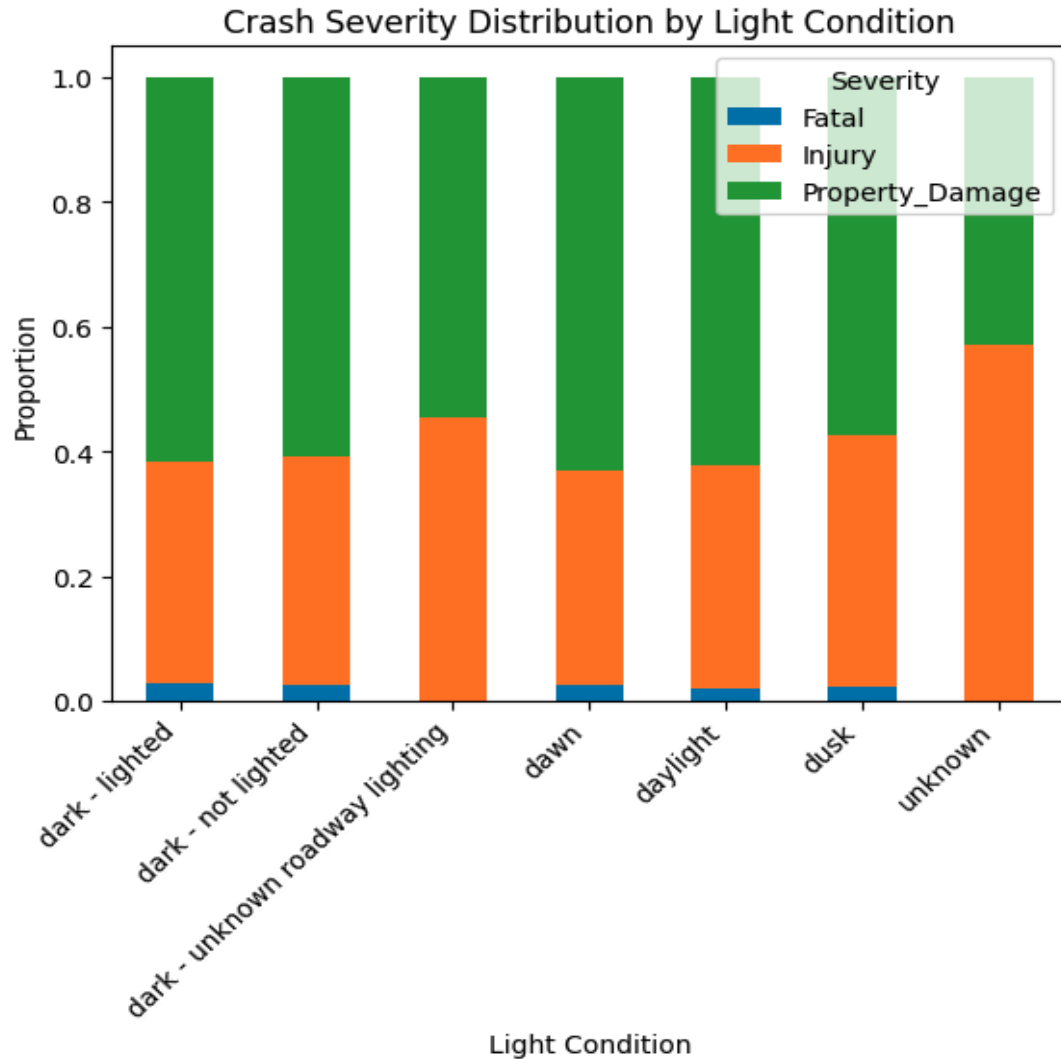
Instead of just predicting crash occurrence, the model captures real-world severity levels (PDO, Injury, Fatal) for better decision-making

Exploratory Visualizations - 2



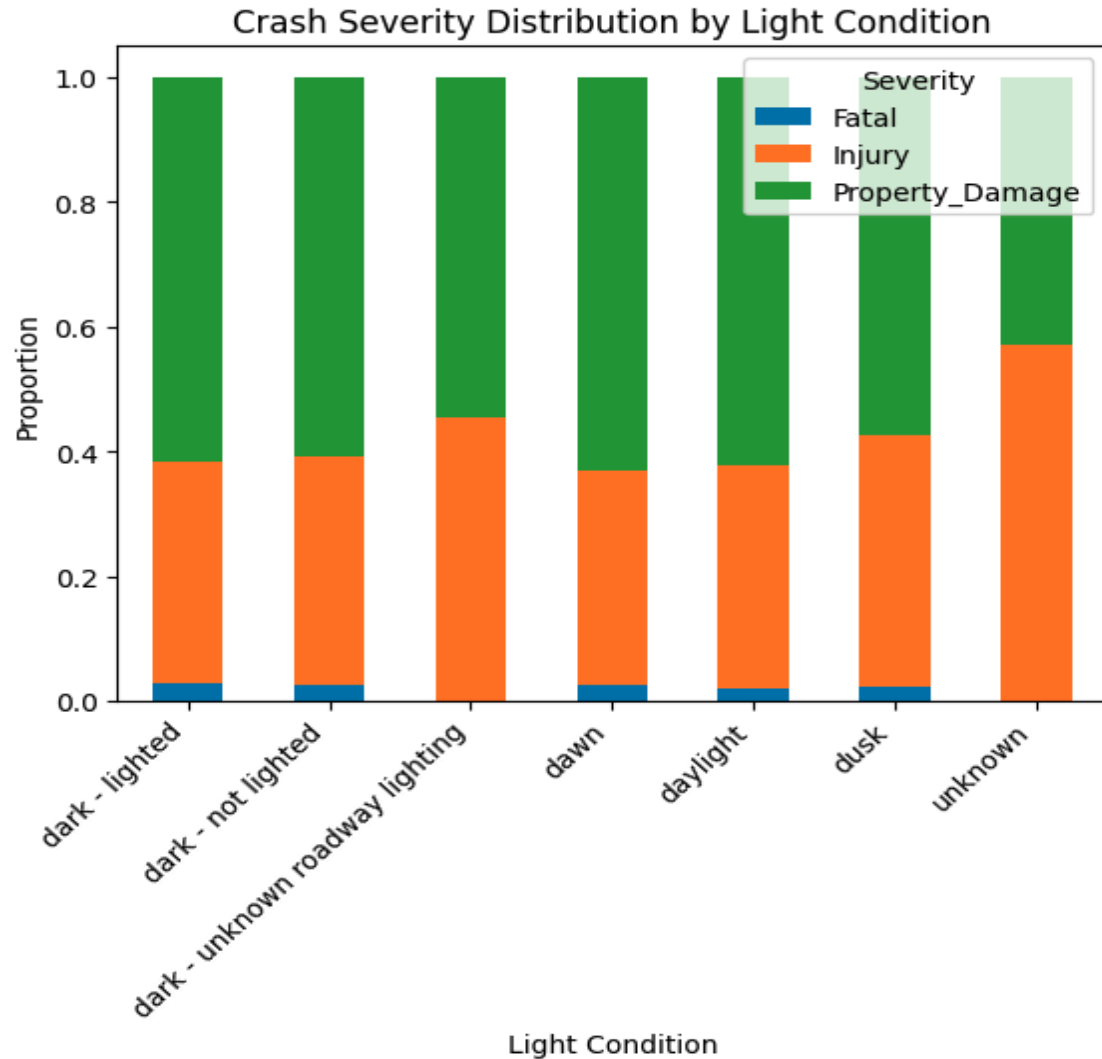
- *Adverse weather increases injury risk*
- *Crashes still occur in normal conditions*
- *Fatal crashes are rare and not weather-specific*

Exploratory Visualizations - 3



- **Nighttime driving increases injury risk**
- **Lighting helps—but doesn't eliminate risk**
- **Daylight is safest; light condition is a key predictor**
- **Daytime has more property-damage-only crashes**

Exploratory Visualizations - 4



- **Slippery surfaces increase injury risk**
- **Dry roads are relatively safer**
- **Severe conditions show mixed behavior**

EDA Summary

Overall Findings:

Across the analyses:

 Weather **Moderate** effect → impacts injury risk

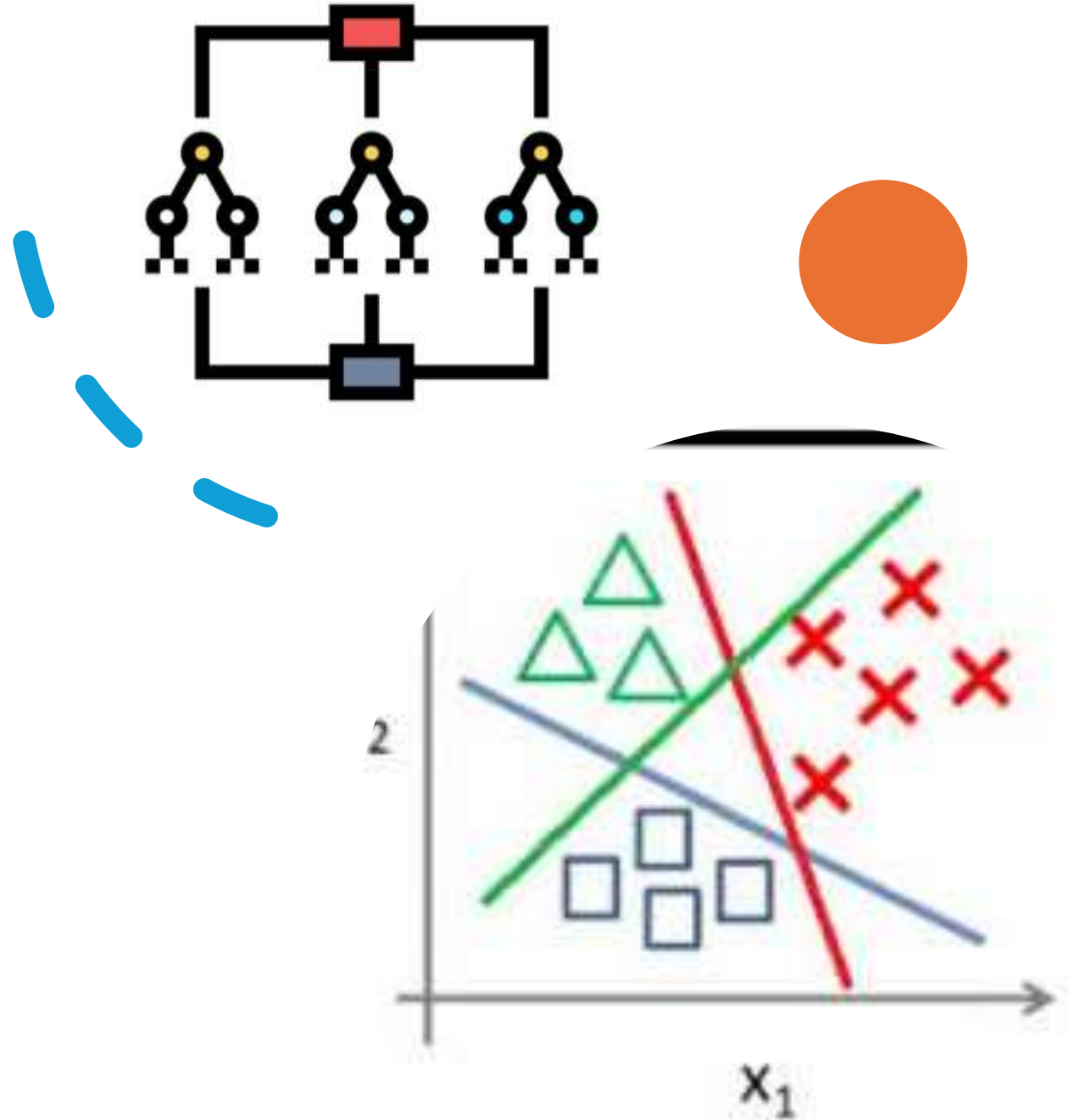
 Light **Condition** Strongest effect → visibility-driven severity

 Road Surface **Moderate** effect → traction-driven severity

Model Building and Testing

Models Tested :

- *Multiclass Logistic Regression - Baseline Model*
- *Multiclass Logistic Regression - Class Weighting*
- *Random Forest*
- *Random Forest with combined Features - Final Model*



Model Comparison



Model Comparison Report

Model Name	Accuracy	Precision (Avg)	Recall (Avg)	F1-Score (Avg)	Minority Recall (Fatal/Injury)	Remarks
Logistic Regression (Baseline)	0.62	0.38	0.62	0.47	0.00 / 0.00	High accuracy driven by majority class; fails completely to detect severe crashes
Logistic Regression (Class Weighting)	0.26	0.50	0.26	0.30	0.50 / 0.40	Improves detection of severe crashes but sacrifices overall accuracy significantly
Random Forest	0.20	0.56	0.20	0.27	0.80 / 0.14	Strong detection of fatal crashes but poor balance across other classes
Random Forest (Final – Combined Features)	0.68	0.52	0.48	0.49	0.60 / 0.42	Best balance between accuracy and severe crash detection; most practical for real-world use

The Product

Web Application deployed on Hugging face -

<https://huggingface.co/spaces/TommyT777/UPS-Crash-Risk>

Let's do a live demo

Spaces

TommyT777/UPS-Crash-Risk

like 0

Agents

Running

Logs

App

Files

Community

Settings

UPS Delivery Crash Risk Estimator

This app estimates crash severity risk using the final Random Forest workflow from the modeling notebook. Inputs were simplified into driver-facing categories for weather, light, road surface, trafficway type, and hour of day. Important limitation: this model reflects the uploaded SMS crash dataset and has low overall predictive accuracy, despite stronger recall for fatal cases.

Weather

clear

Light Condition

day

Road Surface

dry

Trafficway Type

divided

Hour of Day

0

12

23

Clear

Submit

Primary Predicted Severity

Risk Summary and Guidance

Predicted Probability Breakdown

1	2	3
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Share via Link

Limitations



Corporate Roadblocks!

The biggest limitation here is not the model, it's the data. If we had access to driver behavior, route exposure, and operational variables, this would move from a prototype to a production-level system.

- *Lack of Operational & Driver-Level Data*
- *No Route-Level or Exposure Data*
- *Limited Predictive Depth Due to Data Constraints*
- *Lack of driver compensation*

Business Impact and Conclusions

Business Impact

- The driver can assess risk based on core physical conditions
- Gain perspective on how severe the trip is
- Set expectations based on individual driving skills as well

Conclusion

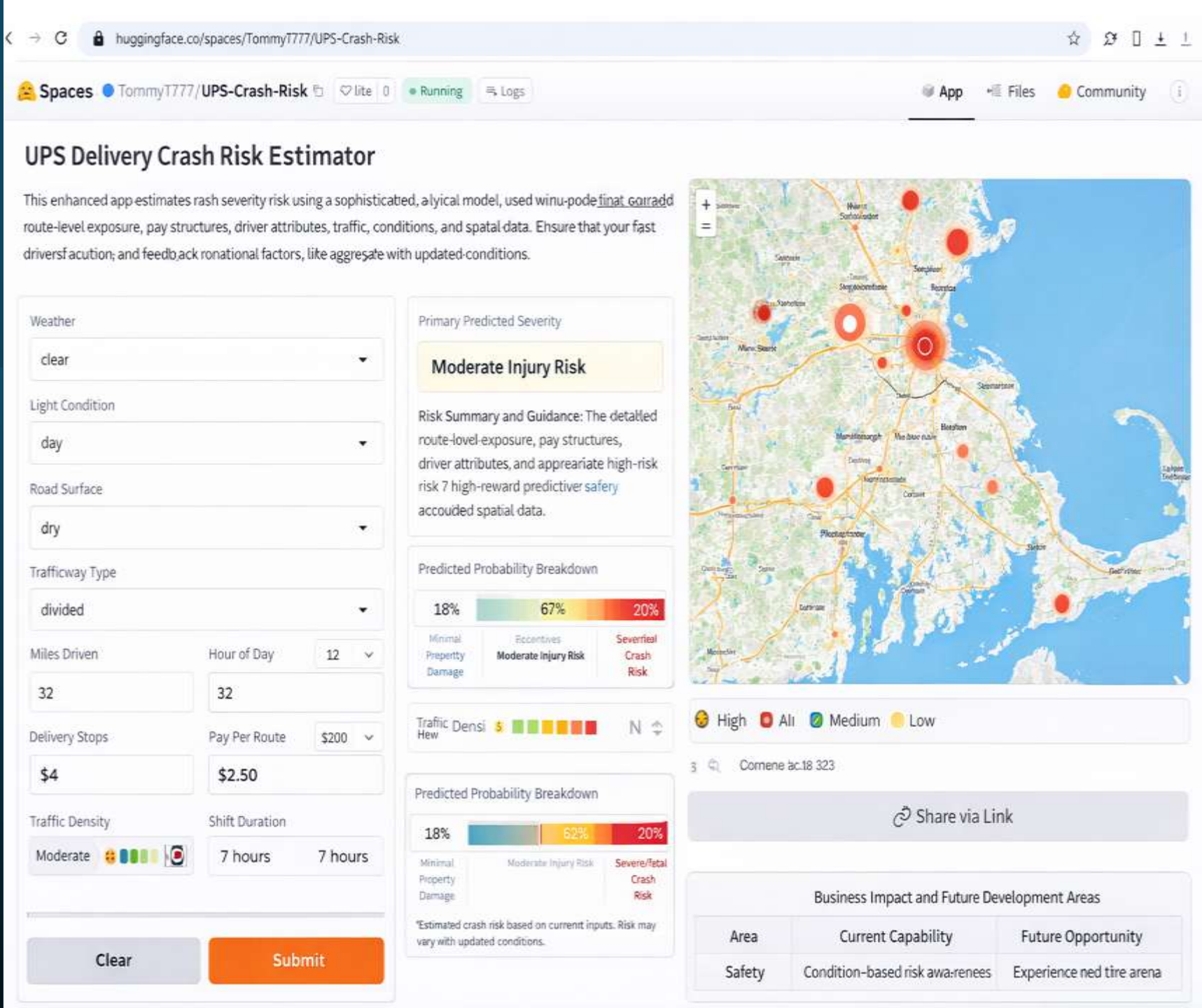
- Developed a data-driven crash risk framework
Successfully modeled crash severity using real-world data and translated outputs into a driver-friendly decision-support tool
- Key insight: severity is context-driven
Factors like weather, light, and road conditions significantly influence injury risk, while fatal events remain rare but critical
- Future potential
With access to richer operational data, this system can evolve into a real-time, high-precision safety platform for delivery operations

Future steps (The dream)

If only we had:

- More enriched data
- More parameters
- Real time data acquisition

AI Generated
Image



Questions